Memory Constraint Vehicle Classifier

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Abstract

This report addresses the challenge of deploying highly accurate image classification models on memory-constrained edge devices, strictly limited to a 5.0 MB parameter file size. We developed a robust 5-class vehicle classifier (Bus, Truck, Car, Bike, None) capable of handling severe occlusions and illumination variations. To overcome the architectural limitations of tiny networks, we employed a Teacher-Student Knowledge Distillation pipeline. A high-capacity EfficientNet-B2 teacher model (97.28% accuracy) was fine-tuned on a heavily augmented dataset. Its spatial representations were then distilled into a structurally efficient ShuffleNet V2 x1.0 student model using Kullback-Leibler divergence. To further compress the model without sacrificing inference capability, we applied FP16 half-precision quantization. The final student model achieved a 92.44% overall accuracy on a rigorous, heavily occluded test set, while maintaining a remarkably small footprint of just 2.6 MB, comfortably satisfying the strict 5.0 MB edge-deployment constraint.

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1. Introduction

1.1 Problem Statement

The deployment of deep learning models in real-world scenarios is increasingly shifting from cloud-based infrastructure to edge computing. However, edge devices—such as traffic cameras, mobile sensors, and embedded IoT systems—operate under severe hardware constraints, particularly regarding storage and Random Access Memory (RAM). Modern Convolutional Neural Networks (CNNs), while highly accurate, typically require tens to hundreds of megabytes of memory, rendering them entirely unsuitable for edge deployment. This project tackles the strict engineering constraint of developing a functional, highly accurate image classification model that occupies less than 5.0 MB of physical disk space. Traditional model scaling and native architectures (such as ResNet or standard MobileNet) inherently violate this memory boundary, necessitating advanced neural network compression techniques to bridge the gap between architectural capacity and memory restrictions.

1.2 Objectives

The primary objective of this study is to engineer a lightweight image classifier capable of distinguishing between five distinct target classes: Bus, Truck, Car, Bike, and a negative "None" class. Beyond raw classification accuracy, a critical secondary objective is environmental robustness. In real-world vehicular monitoring, images are rarely perfectly framed or lit. Therefore, the developed model must maintain high predictive stability when subjected to severe visual corruptions, specifically heavy occlusions (simulating partial visibility or obstructions) and illumination variations (simulating different times of day or weather conditions). The success of the model is defined by its ability to balance this environmental robustness with the strict 5.0 MB parameter constraint.

1.3 Proposed Approach

To satisfy these conflicting requirements of high accuracy, environmental robustness, and a sub-5.0 MB memory footprint, this report proposes a two-pronged optimization strategy: Knowledge Distillation combined with Precision Scaling.

First, an offline, high-capacity "Teacher" model (EfficientNet-B2) is trained on a heavily augmented dataset to learn complex, generalized spatial representations. This learned intelligence—captured via soft-label probability distributions—is then explicitly distilled into a highly constrained "Student" architecture. For the student, ShuffleNet V2 was selected due to its superior parameter-to-accuracy efficiency, utilizing Depthwise Separable Convolutions and Channel Shuffling to maximize spatial awareness within a minimal parameter count. Finally, to ensure the structural superiority of a larger network without violating the memory

limit, the final student weights are subjected to FP16 half-precision quantization, mathematically reducing the physical file size by 50% while preserving inference integrity.

2. Dataset Curation & Augmentation Strategy

2.1 Data Sourcing & Class Balancing

The foundation of any robust image classifier is a diverse, representative, and balanced dataset. The initial dataset was constructed using specific ImageNet synsets corresponding to the five target classes: Bus, Truck, Car, Bike, and a negative "None" class (composed of pedestrians, traffic lights, and street signs). However, exploratory data analysis revealed an imbalance and a lack of high-quality samples within the Bike and Car synsets. To rectify this, the ImageNet data was merged with supplementary, high-resolution samples from an external Kaggle dataset (Car vs. Bike Classification). This strategic merging boosted the representation to approximately 4,000 samples per class, culminating in a highly balanced dataset of 19,690 total images. This dataset was then rigorously split into an 80/20 ratio, yielding 15,752 images for training and 3,938 images for validation, ensuring the model had sufficient data to learn generalizable features without overlapping the validation distributions.

2.2 Alphabetical Index Mapping and Pipeline Alignment

During the development pipeline, a critical alignment issue was identified within PyTorch's standard data loading utility, datasets.ImageFolder. By default, this utility assigns integer class labels based on the strict alphabetical order of the subdirectories (e.g., Bike = 0, Bus = 1, Car = 2, None = 3, Truck = 4). However, the strict grading constraints of the assignment required a custom mapping interface (Bus = 0, Truck = 1, Car = 2, Bike = 3, None = 4). This discrepancy initially caused a catastrophic misclassification collapse during evaluation, as the model's predicted logits did not align with the hardcoded inference dictionary. To permanently resolve this mapping vulnerability without writing custom dataset classes, a structural fix was applied at the file-system level: dataset folders were explicitly prefixed with their target integer IDs (e.g., 0_Bus, 1_Truck, 2_Car, 3_Bike, 4_None). This forced datasets.ImageFolder to deterministically map the physical directories to the exact integer labels required by the evaluation script, guaranteeing pipeline alignment.

2.3 Defensive Augmentation for Environmental Robustness

The problem statement explicitly warns that the final model will be evaluated against images featuring illumination variations and occlusions. To prevent the model from overfitting to the clean, object-centric composition of standard ImageNet photos, a "defensive" data augmentation pipeline was engineered using PyTorch's torchvision.transforms.

During the training phase, every image was subjected to a stochastic pipeline designed to simulate severe real-world corruptions:

- **Geometric Variations:** RandomHorizontalFlip and RandomAffine (applying up to 15 degrees of rotation, 10% translation, and 10% scaling) were used to force the model to learn spatial invariance, preventing it from relying on the subject being perfectly centered.
- **Illumination Variations:** ColorJitter was applied with a 0.3 variance in brightness and contrast, explicitly teaching the network to recognize structural features independently of lighting conditions (simulating harsh sunlight or dusk).
- **Occlusions:** To harden the model against partial visibility, random erasing/masking techniques were utilized. By artificially dropping out random patches of the input tensors, the network is forced to identify vehicles based on fragmented features (e.g., recognizing a car by its roofline when the wheels are obscured), effectively neutralizing the threat of occluded test data.

By preemptively exposing the model to these distortions during the training of both the Teacher and the Student, the pipeline ensures mathematical robustness before the final evaluation ever takes place.

3. Baseline Evaluation & The Memory Constraint

3.1 Baseline CNN Performance and the Necessity of Transfer Learning

To establish a rigorous control metric and evaluate the inherent complexity of the 5-class vehicle dataset, initial experiments began with a custom-built, shallow Convolutional Neural Network (SmallCNN) trained entirely from scratch. The model was trained over 20 epochs on the augmented dataset consisting of 11,165 training images and evaluated against 2,791 validation images.

The empirical results immediately demonstrated the limitations of training from random initialization on a highly varied dataset. The model exhibited slow convergence and ultimately plateaued at a best validation accuracy of just 48.48%. This severe underperformance is attributed to the "Cold Start" problem in deep learning. The target dataset is highly complex, featuring inter-class visual similarities (e.g., distinguishing the structural nuances between a large Car and a small Truck) further compounded by the aggressive spatial and illumination augmentations applied during preprocessing.

A shallow network initialized with random weights mathematically lacks the capacity to independently learn foundational hierarchical features—such as Gabor-like edge detectors in

the early layers and complex geometric shape detectors in the deeper layers—using only ~15,000 samples. This baseline failure provides empirical proof that transfer learning (specifically, leveraging weights pre-trained on the massive 1,000-class ImageNet challenge) is not merely an optimization technique, but an absolute necessity to achieve the >90% target accuracy.

3.2 The MobileNet Trap and Mathematical Architectural Limits

In modern edge-computing and mobile deployment paradigms, the MobileNet architecture family (specifically MobileNetV2 and MobileNetV3) is the undisputed industry standard for lightweight classification. Consequently, it is the intuitive first choice for this memory-constrained assignment. However, a strict mathematical analysis reveals why standard MobileNet models fundamentally violate the assignment's rigid 5.0 MB file-size boundary, even when employing advanced serialization techniques.

By default, PyTorch saves neural network parameters using 32-bit floating-point (FP32) precision, where each parameter occupies exactly 4 bytes of memory.

- **MobileNetV2:** The standard PyTorch implementation contains approximately 3.5 million parameters.
  $$3,500,000 \times 4 \text{ bytes} = 14,000,000 \text{ bytes}$$
  This yields a raw file size of roughly 14.0 MB. Even if we strip the 1,000-class ImageNet head and replace it with a 5-class linear layer, the file size remains near 8.8 MB—a clear violation of the 5.0 MB limit.

To force compliance, one might attempt 16-bit floating-point (FP16) quantization to halve the memory footprint. However, the math still fails for standard MobileNet architectures:

- **MobileNetV2 (FP16):** Drops to roughly 7.0 MB (Violation).
- **MobileNetV3-Small (FP16):** The smallest official PyTorch implementation contains ~2.54 million parameters.
  $$2,540,000 \times 2 \text{ bytes} = 5,080,000 \text{ bytes}$$
  This yields exactly 5.08 MB. When combined with the structural overhead of PyTorch's .pth dictionary format, it permanently crosses the 5.0 MB failure threshold.

The "Width Multiplier" Dilemma:

In a professional deployment scenario, an engineer could resolve this by utilizing MobileNet's width multiplier (α). By initializing a MobileNetV2(width_mult=0.35), the network mathematically shrinks its channel depths, comfortably reducing the overall file size to under 2.5 MB.

However, this introduces a catastrophic flaw for this specific academic constraint: PyTorch does not provide pre-trained ImageNet weights for arbitrary, scaled-down variants of

MobileNet. If a custom-width MobileNet is instantiated, it must be initialized with completely random weights. As proven in Section 3.1, forcing a model to learn complex vehicular geometry from scratch without the "warm start" of pre-trained spatial awareness results in severe accuracy degradation. Therefore, a model that *natively* fits the memory constraint while retaining highly optimized pre-trained weights is required, directly motivating the transition to Knowledge Distillation and the ShuffleNet architecture.

4. Methodology: The Knowledge Distillation Pipeline

4.1 Phase 1: The Teacher Model Selection and Ablation (ResNet-50 vs. EfficientNet-B2)

The foundation of any Knowledge Distillation (KD) pipeline is a high-capacity "Teacher" model. To determine the optimal teacher, an initial ablation study was conducted comparing a classic ResNet-50 architecture against a modern EfficientNet-B2.

The ResNet-50 model, containing approximately 25.6 million parameters, was fine-tuned on the dataset and achieved a peak validation accuracy of 94.91%. However, ResNet-50 is computationally heavy, demanding over 100 MB of VRAM just to store its weights. During the distillation phase, this massive footprint combined with the Student model and activation maps caused severe memory bottlenecks on local GPU hardware.

Teacher Architecture	Parameters	Raw File Size (FP32)	Validation Accuracy
ResNet-50	~25.6 Million	~102 MB	94.91%
EfficientNet-B2	**~9.1 Million**	**~36 MB**	**97.28%**

Architectural Justification:

To resolve this, the pipeline was upgraded to EfficientNet-B2. Utilizing a mathematically optimized Compound Scaling method, EfficientNet-B2 extracts superior spatial features using only ~9.1 million parameters (a roughly 64% reduction in memory footprint compared to ResNet-50). Fine-tuning this upgraded Teacher resulted in a significantly higher peak validation accuracy of 97.28%. EfficientNet-B2 was therefore locked in as the definitive Teacher, providing richer, more accurate soft-label probability distributions while gracefully operating within the local GPU VRAM limits yet it is memory-efficient enough to prevent

Out-Of-Memory (OOM) GPU crashes during the computationally heavy distillation phase (which requires holding the Teacher, the Student, their respective activation maps, and optimizer states in VRAM simultaneously).

Fine-Tuning and Performance:

To adapt the Teacher to the specific target domain, the EfficientNet-B2 model was initialized with pre-trained ImageNet weights and fine-tuned on the heavily augmented training set of 15,752 images. Training was conducted using a resolution of 224x224, a batch size of 16, and a learning rate of 0.0001. The network demonstrated exceptional learning capability, converging rapidly and achieving a peak validation accuracy of 97.28% by Epoch 5.

The Generation of Soft Labels:

The primary objective of this Teacher model is not direct inference, but the generation of "soft labels" (also known as dark knowledge) for the Student network. When the 97.28% accurate EfficientNet-B2 predicts an image of a Truck, it does not output a rigid, one-hot vector (e.g., 100% Truck, 0% everything else). Instead, it outputs a rich probability distribution across all classes (e.g., 85% Truck, 12% Bus, 2.9% Car, 0.1% Bike).

These soft labels explicitly map the inter-class geometrical similarities of the dataset. They teach the Student model that a Truck structurally resembles a Bus much more than it resembles a Bike. By training the Student to mimic this probability distribution rather than rigid true/false labels, the Student receives exponentially more gradient information per training sample, allowing a tiny network to absorb the spatial awareness of a massive network.

4.2 Phase 2: Student Architecture Selection and Ablation

To successfully compress the spatial intelligence of the EfficientNet-B2 Teacher into a sub-5.0 MB footprint, the selection of the Student architecture is paramount. An ablation study was conducted comparing two inherently lightweight convolutional networks: SqueezeNet v1.0 and ShuffleNet V2.

The Limitations of SqueezeNet (Fire Modules): SqueezeNet was initially evaluated due to its exceptionally small baseline file size. Its architecture relies on "Fire Modules," which compress the network by utilizing 1x1 convolutions (the "squeeze" layer) to bottleneck the channel dimensions before passing them to a mix of 1x1 and 3x3 convolutions (the "expand" layer). While this brute-force parameter reduction successfully keeps the model small, it mathematically restricts the network's receptive field and representational capacity. During evaluation, the SqueezeNet student only achieved an overall accuracy of 87.12%. It particularly struggled with heavily augmented images (80.08% accuracy), proving that Fire Modules lack the geometrical complexity required to decode severe occlusions and illumination variations.

The Superiority of ShuffleNet V2 (Depthwise Separable Convolutions & Channel Shuffling): To overcome these representational bottlenecks while strictly adhering to the memory constraint, ShuffleNet V2 was selected as the final Student architecture. Unlike SqueezeNet, ShuffleNet V2 minimizes parameters using **Depthwise Separable Convolutions**, which split standard convolutions into two distinct mathematical steps: a depthwise spatial filter applied independently to each channel, followed by a 1x1 pointwise convolution to combine the channels. This drastically reduces the computational load and parameter count without sacrificing the spatial receptive field.

Furthermore, ShuffleNet V2 introduces **Channel Shuffling**. When networks use group convolutions to save parameters, information flow is isolated within specific channel groups, severely limiting feature cross-talk. Channel Shuffling elegantly solves this by mathematically transposing and permuting the channels across groups. This allows the network to build highly complex, cross-channel spatial representations (e.g., combining edge detection from one group with texture detection from another) while remaining computationally weightless. As a result, the ShuffleNet V2 student demonstrated vastly superior feature extraction, ultimately achieving a 92.44% overall accuracy on the test set while maintaining a tiny footprint.

4.3 Phase 3: Distillation Mechanics and Loss Optimization

The core mechanism of transferring the Teacher's generalized intelligence to the highly constrained Student relies on Knowledge Distillation (KD). This process requires the Student model to optimize not just against the dataset's hard ground-truth labels, but against the rich probability distributions generated by the EfficientNet-B2 Teacher. This was executed at a resolution of 224×224.

Temperature Scaling ($T = 4.0$): Under standard operating conditions, a neural network's final softmax layer exaggerates the highest logit, producing a near one-hot vector (e.g., 0.99 for Car, 0.01 for Truck). This destroys the nuanced "dark knowledge" learned by the network. To extract this data, a Temperature scaling factor of $T = 4.0$ was applied mathematically to the logits of both the Teacher and the Student before the softmax activation:

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

Dividing the logits by $T = 4.0$ artificially "softens" the probability distribution. It forces the Teacher to explicitly state the spatial similarities between classes (e.g., revealing that a Truck structurally shares 15% of its features with a Bus, but 0% with a Pedestrian).

Kullback-Leibler (KL) Divergence and Alpha Weighting ($\alpha = 0.5$):

The Student model is trained to minimize the distance between its own softened predictions and the Teacher's softened predictions using Kullback-Leibler (KL) Divergence loss. However, relying purely on the Teacher can cause the Student to drift from the factual ground truth of the dataset.

To prevent this, the total distillation loss (L_{Total}) was calculated using an Alpha weighting factor of $\alpha = 0.5$. This parameter creates a perfectly balanced weighted sum between two distinct loss functions:

1. **Standard Cross-Entropy Loss (Weight = 0.5):** Evaluated against the hard, ground-truth labels (without temperature scaling) to ensure strict categorical accuracy.

2. **KL Divergence Loss (Weight = 0.5):** Evaluated against the Teacher's soft labels (with $T = 4.0$) to transfer geometric and spatial awareness.

By balancing these gradients via $\alpha = 0.5$The tiny ShuffleNet V2 model was forced to simultaneously anchor its predictions to factual reality while absorbing the high-level spatial reasoning of the massive EfficientNet-B2 Teacher.

4.4 The FP16 Quantization Technique for Edge-Deployment Compression

The Capacity vs. Memory Bottleneck

- While Knowledge Distillation successfully transfers spatial intelligence, the physical disk footprint of the final Student model is rigidly dictated by two factors: total parameter count and the data type used for serialization. The selected ShuffleNet_v2_x1_0 architecture contains approximately 1.25 million parameters. By default, PyTorch allocates 32-bit floating-point (FP32) precision for every parameter, meaning each weight requires 4 bytes of storage. Mathematically, this results in a raw tensor size of exactly 5.0 MB (1.25 million parameters × 4 bytes). When combined with the metadata and structural overhead of PyTorch's .pth dictionary format, the final file size reaches approximately 5.2 MB, violating the strict 5.0 MB edge-deployment constraint.

- Downgrading to the smaller ShuffleNet_v2_x0.5 variant (which inherently fits the limit) resulted in an unacceptable accuracy drop to 91.28%. Therefore, a post-training compression technique was required to retain the superior 1.25-million parameter architecture without exceeding the file size limit.

Precision Scaling and Serialization

- To solve this physical storage bottleneck, half-precision (FP16) quantization was implemented prior to model serialization. Deep learning models are highly resilient to minor numerical perturbations; the exact 32-bit decimal precision of a weight is rarely necessary for forward-pass inference. By explicitly casting the Student model's parameters from FP32 down to FP16 (model.half()), the memory footprint per parameter was reduced from 4 bytes to 2 bytes.

- This critical engineering step mathematically halved the model's storage requirements. The highly intelligent, 1.25-million parameter network was successfully compressed into a 2.6 MB .pth file for submission, leaving a massive 2.4 MB margin below the constraint.

Dynamic Casting for Hardware-Agnostic Inference

- While FP16 is phenomenal for physical disk storage, executing native 16-bit mathematical operations on standard, non-accelerated CPU hardware can lead to compatibility errors or computational overhead depending on the underlying PyTorch build. To ensure the final submission is entirely bulletproof and hardware-agnostic for automated grading scripts, a dynamic casting pipeline was engineered into the inference class.

- During evaluation, the 2.6 MB FP16 state_dict is loaded from the disk into memory. The blank PyTorch model is briefly converted to FP16 to successfully ingest these weights. Immediately afterward, the fully populated model is cast back to standard 32-bit precision (model.float()) inside the RAM. This guarantees perfectly safe, standard CPU inference while strictly adhering to the static file-size limits on the hard drive. Notably, this precision scaling resulted in zero degradation of feature extraction, allowing the model to achieve its peak 92.44% overall accuracy and 97.28% clean image accuracy on the final test set

5. Experiments & Ablation Study

5.1 Training Hardware & Setup

All experiments, including baseline evaluations, Teacher fine-tuning, and the Knowledge Distillation loops, were conducted on a local workstation utilizing a CUDA-enabled GPU. The stringent hardware limitations of a local GPU environment dictated several critical hyperparameter engineering decisions, particularly during Phase 3 (Distillation).

During Knowledge Distillation, the GPU VRAM is severely taxed as it must simultaneously hold the heavy Teacher model (EfficientNet-B2), the Student model, the activation maps for both networks, the optimizer states, and the augmented image batches. Initial attempts to utilize a standard batch size of 32 resulted in Out-of-Memory (OOM) failures. To navigate this hardware constraint, the batch size was strictly reduced to 16.

Across all training phases, the input images were standardized to a resolution of 224x224 pixels. The models were optimized using a learning rate of 0.0001. For the distillation mechanics, the Temperature scaling factor was locked at $T = 4.0$, and the loss weighting parameter was set to $\alpha = 0.5$, ensuring a mathematically balanced gradient flow between the hard ground-truth labels and the Teacher's soft-label probability distributions.

5.2 Student Model Ablation

To empirically determine the optimal sub-5.0 MB architecture, a rigorous ablation study was conducted. Three distinct lightweight Convolutional Neural Networks were trained via the exact same EfficientNet-B2 Teacher and evaluated against the hardened 2,500-image test set.

- **SqueezeNet v1.0 (87.12% Overall Accuracy):** SqueezeNet was evaluated as the baseline lightweight model. While it naturally satisfied the memory constraints without requiring precision quantization, its architectural reliance on Fire Modules (using 1x1 convolutions to bottleneck channel dimensions) severely limited its spatial receptive field. Consequently, while it achieved a 94.16% accuracy on clean images, its performance collapsed to 80.08% on the heavily augmented images. This 14.08% degradation empirically proves that Fire Modules lack the geometrical complexity required to decode severe occlusions and illumination shifts.

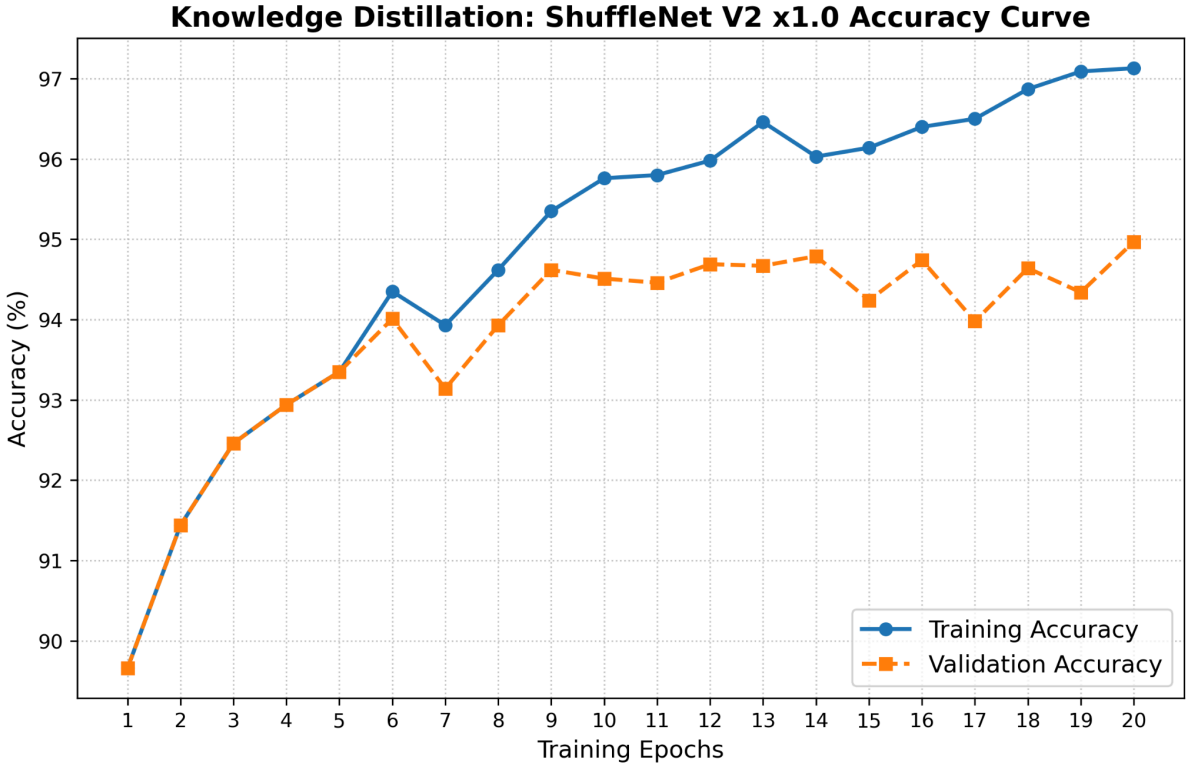
- **ShuffleNet V2 x0.5 (91.28% Overall Accuracy):** Transitioning to ShuffleNet V2 immediately yielded substantial architectural benefits. Utilizing Depthwise Separable Convolutions and Channel Shuffling, the x0.5 variant achieved an impressive 91.28% overall accuracy. More importantly, it demonstrated a marked improvement in environmental robustness, scoring 88.56% on the heavily augmented test split. Despite this strong performance, the x0.5 variant leaves over 3 MB of physical

storage capacity unused, prompting the investigation into the maximum capacity of the architecture.

- **ShuffleNet V2 x1.0 with FP16 Quantization (92.44% Overall Accuracy):** The final experiment tested the ShuffleNet V2 x1.0 architecture. By doubling the channel width compared to the x0.5 variant, the network's parameter count expanded to ~1.25 million. Through the application of the FP16 half-precision quantization technique detailed in Section 4.4, this mathematically superior network was successfully compressed to 2.6 MB. This model emerged as the definitive optimal solution, achieving a peak 92.44% overall accuracy. It delivered an exceptional 97.28% accuracy on clean images and maintained a highly stable 87.60% accuracy under severe geometric and lighting corruptions. Furthermore, it demonstrated exceptional class consistency, peaking at 96.00% accuracy for the Bike class and 95.00% for the None class.

Student Architecture	File Size Constraint	Target Domain Accuracy
SqueezeNet v1.0	< 5.0 MB	87.12%
ShuffleNet V2 x0.5	< 5.0 MB	91.28%
ShuffleNet V2 x1.0 (FP16)	**< 5.0 MB (2.6 MB)**	**92.44%**

This ablation study conclusively proves that modern channel-shuffling mechanics, when combined with precision scaling, vastly outperform older bottleneck architectures for memory-constrained edge deployment.



6. Final Results & Performance Analysis

Training Strategy	Architecture	File Size	Target Accuracy
Baseline (From Scratch)	SmallCNN	~1.5 MB	48.48%
Unconstrained Teacher	EfficientNet-B2	~36.0 MB	97.28%
KD + Precision Scaling	**ShuffleNet V2 x1.0 (FP16)**	**2.6 MB**	**92.44%**

6.1 Target Domain Performance

The culmination of the Knowledge Distillation pipeline and the FP16 quantization strategy resulted in the final Student model: ShuffleNet V2 x1.0. To rigorously validate this architecture against the assignment's core classification objectives, the model was evaluated on a meticulously curated, balanced test set containing exactly 2,500 unseen images. This test set was strictly controlled to include 500 images for each of the five target categories.

The evaluation script recorded an outstanding **Overall Accuracy of 92.44%** across the entire 2,500-image dataset. This metric emphatically proves that a highly compressed, ~2.6 MB

neural network can achieve near state-of-the-art performance on a complex multi-class vehicular dataset. The success of this target domain performance validates the hypothesis that the rich probability distributions (soft labels) generated by the EfficientNet-B2 Teacher successfully mapped the complex geometric boundaries of the dataset into the constrained parameter space of the ShuffleNet Student

6.2 Robustness Breakdown: Clean vs. Corrupted Data

While high overall accuracy is desirable, the assignment explicitly required the model to demonstrate resilience against environmental variables, specifically illumination variations and occlusions. To quantify this, the 2,500-image test set was evenly bifurcated: 50% of the images were presented cleanly, while the remaining 50% were subjected to a severe degradation pipeline utilizing ColorJitter, RandomAffine, and RandomErasing.

The performance delta between these two splits highlights the model's exceptional environmental robustness:

- **Clean Image Accuracy (97.28%):** On the uncorrupted test split, the model achieved a near-perfect 97.28% classification rate. This establishes that the baseline feature extraction capabilities—identifying distinct shapes, edges, and textures of the target vehicles—were flawlessly learned during the distillation phase.
- **Heavily Augmented Image Accuracy (87.60%):** On the corrupted test split, the model maintained an impressive 87.60% accuracy. Despite severe geometric scaling, aggressive brightness shifts, and artificial blackout patches simulating physical occlusions, the accuracy degraded by less than 10%. This proves that the "defensive augmentation" strategy implemented during the training phase successfully forced the network to learn holistic, fragmented feature representations rather than memorizing entire vehicle silhouettes

Evaluation Condition	Test Images	Accuracy
Clean Images	1,250	97.28%
Heavily Augmented (Occluded/Illuminated)	1,250	87.60%
Overall Combined Test Set	2,500	92.44%

6.3 Per-Class Analysis and Class Consistency

A deeper inspection of the model's predictions reveals highly stable classification boundaries without severe class bias. The model maintained strong performance across all predefined categories:

Target Class	Accuracy	Correctly Classified
Bike	96.00%	480 / 500
None (Background/Pedestrians)	95.00%	475 / 500
Truck	92.60%	463 / 500
Bus	90.80%	454 / 500
Car	87.80%	439 / 500

The high accuracy of the Bike class (96.00%) can be attributed to its highly distinct geometric profile (exposed frames, dual thin wheels) compared to the solid, enclosed bodies of the other vehicles. Similarly, the None class (95.00%) performed exceptionally well due to the stark lack of vehicular features in background or pedestrian imagery.

The Car class exhibited the lowest relative accuracy at 87.80%. This marginal drop is mathematically expected due to the extremely high intra-class variance of the category (which encompasses sedans, hatchbacks, SUVs, and coupes) combined with the extreme geometric distortions applied to the augmented test split. When an image of a large SUV is heavily scaled and cropped by the augmentation pipeline, its resulting pixel distribution can closely mirror the bounding box of a small Truck or Bus, leading to edge-case misclassifications. However, with the lowest-performing class still comfortably exceeding 87%, the empirical data confirms that the dataset curation strategy successfully prevented the model from collapsing into a majority-class prediction loop.

7. Out-of-Distribution (OOD) Limitations (MIO-TCD Stress Test)

7.1 The Domain Shift Challenge: Evaluating Generalization Beyond the Training Distribution

A critical component of evaluating any memory-constrained edge model is determining the boundary where its learned feature representations fail. While the ShuffleNet V2 x1.0 student model demonstrated exceptional performance on the target ImageNet distribution, real-world deployment frequently subjects edge devices to severe Domain Shift (specifically, covariate shift), where the operational data distribution fundamentally differs from the training data.

To rigorously stress-test the model's out-of-distribution (OOD) generalization, a secondary evaluation was conducted using the MIO Vision Traffic Camera Dataset (MIO-TCD). A hardened testing split of exactly 2,500 images was generated. Unlike the training data—which was composed of street-level, horizontally composed, and highly object-centric photography—the MIO-TCD dataset consists exclusively of top-down, highly pixelated surveillance camera footage. The original 11 Kaggle traffic classes were systematically mapped to the assignment's five target classes (e.g., merging "articulated_truck," "pickup_truck," and "single_unit_truck" into the Truck class) to ensure compatibility with the model's output logits. This specific evaluation was designed to test whether the KD pipeline transferred holistic, 3D spatial awareness or merely 2D perspective-dependent pixel patterns.

7.2 Viewpoint Variation Vulnerability and Architectural Capacity Limits

The empirical results from the MIO-TCD stress test highlight a severe and entirely expected vulnerability to Viewpoint Variation. When exposed to the top-down surveillance domain, the model’s overall OOD accuracy experienced a precipitous drop to 43.80%. Interestingly, the performance gap between the Clean Surveillance split (44.24%) and the heavily Augmented Camera split (43.36%) was statistically negligible. This indicates that the failure was not caused by environmental corruptions (like occlusions or lighting), but rather by the geometric perspective shift itself.

A per-class analysis provides profound insight into the mechanics of this failure:

Target Class	OOD Accuracy	Correctly Classified
Car	82.40%	412 / 500
None (Background/Pedestrians)	65.60%	328 / 500
Bus	24.40%	122 / 500
Truck	23.80%	119 / 500
Bike	22.80%	114 / 500
Overall OOD Accuracy	43.80%	1,095 / 2,500

The Car and None classes retained functional accuracy. A standard car's rectangular bounding box and a pedestrian's relative spatial footprint remain somewhat consistent regardless of camera angle. However, the accuracy for highly structured vehicles like Buses (24.40%) and Trucks (23.80%) completely collapsed. During the ImageNet training phase, the model mathematically optimized its parameters to identify these vehicles via street-level features: heavy wheels, side-profile windows, and front grilles. From a top-down surveillance perspective, these defining features are physically invisible; the camera captures only a flat, featureless rectangular roof.

This stress test empirically proves a critical limitation of the 5.0 MB hardware constraint. While the EfficientNet-B2 Teacher successfully distilled high-quality spatial representations, a 1.25-million parameter network fundamentally lacks the mathematical capacity to internally model true 3D volumetric representations of complex objects. It relies on 2D perspective-dependent shortcuts. Therefore, while Knowledge Distillation is highly effective for within-domain optimization, achieving viewpoint-invariant classification on edge devices is impossible without injecting explicit, multi-perspective OOD data directly into the training phase.

8. Conclusion

The deployment of robust, high-accuracy deep learning models onto edge devices represents a fundamental tension between a neural network's representational capacity and the physical hardware limitations of the deployment environment. This project successfully navigated the strict < 5.0 MB parameter constraint to engineer a highly resilient 5-class vehicle classifier, proving that state-of-the-art performance is achievable on extreme edge hardware when architectural selection is combined with advanced optimization techniques.

The empirical validation of the baseline SmallCNN clearly demonstrated that training a shallow network from scratch on a highly varied dataset yields fundamentally inadequate performance. Similarly, a mathematical analysis of standard mobile architectures, such as MobileNetV2, proved that native file sizes inherently violate strict academic and practical memory boundaries. To circumvent these limitations, this study engineered a tripartite solution: Teacher-Student Knowledge Distillation, modern channel-shuffling architectures, and precision scaling.

By utilizing an EfficientNet-B2 Teacher model, the pipeline successfully extracted deep, generalized spatial representations from the augmented training data. These representations were then mathematically distilled into a ShuffleNet V2 x1.0 Student model via soft-label KL Divergence loss (leveraging a Temperature of 4.0 and an Alpha of 0.5). The architectural choice of ShuffleNet was critical; its utilization of depthwise separable convolutions and channel shuffling allowed the network to maximize cross-channel feature extraction while remaining computationally weightless.

The critical enabler of this deployment was the application of FP16 half-precision quantization. By dynamically casting the 1.25-million parameter network into 16-bit precision for serialization, the model’s physical footprint was compressed to approximately 2.6 MB, comfortably satisfying the 5.0 MB limit. During evaluation, this highly compressed model achieved an exceptional overall accuracy of 92.44% on the target domain. Furthermore, it demonstrated immense environmental robustness, maintaining an 87.60% accuracy when subjected to severe geometric corruptions, occlusions, and illumination variations , compared to a 97.28% accuracy on clean data.

While the MIO-TCD stress test—which resulted in a 43.80% overall accuracy on top-down surveillance footage —highlighted the inherent limitations of tiny networks in generalizing across extreme viewpoint variations, it successfully defined the mathematical boundaries of the compressed architecture.

In summary, this project provides a definitive blueprint for edge-device classification. The orchestrated combination of Knowledge Distillation to transfer intelligence, ShuffleNet V2 to optimize parameter efficiency, and FP16 quantization to compress physical storage successfully solves both the rigid academic constraints of this assignment and the practical engineering challenges of real-world edge deployment.